

## R2 Newton's second law

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time: **234 minutes**

Marks: **192 marks**

Comments: \_\_\_\_\_

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**Q1.**

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force  $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j})$  N acts on the particle for 20 seconds. The unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular and lie in the horizontal plane.

- (a) (i) Find the acceleration of the particle. (2)
- (ii) Find the velocity of the particle at the end of the 20 second period. (2)
- (iii) Find the position of the particle at the end of the 20 second period. (3)
- (b) At the end of the 20 second period, the force  $\mathbf{F}$  is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds. (5)

**(Total 12 marks)**

**Q2.**

A train travels along a straight horizontal track. It starts from rest and accelerates at  $0.5 \text{ m s}^{-2}$  for 10 seconds.

- (a) Find the speed of the train after 10 seconds. (2)
- (b) Find the distance that the train travels in the first 10 seconds. (2)
- (c) The train has a mass of 200 tonnes and experiences a resistance force of 40 000 N. Find the magnitude of the forward driving force that acts on the train while it is accelerating at  $0.5 \text{ m s}^{-2}$ . (3)

**(Total 7 marks)**

**Q3.**

A particle of mass 60 kg is on a rough surface inclined at an angle of  $40^\circ$  to the horizontal.

- (a) Find the magnitude of the normal reaction force acting on the particle. (2)
- (b) If the particle remains at rest, find the minimum value of the coefficient of friction between the particle and the slope. (4)
- (c) If the coefficient of friction between the particle and the slope is 0.2, the particle slides down the slope. Find the acceleration of the particle in this case. (5)

**(Total 11 marks)**

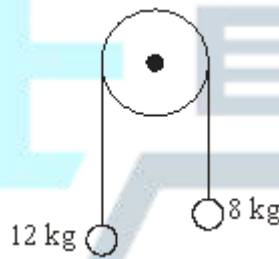
**Q4.**

A block of wood has mass 4 kg. It is placed on a rough horizontal surface and is pulled by a horizontal string. The coefficient of friction between the block and the surface is 0.4.

- (a) Draw a diagram to show the forces acting on the block. (1)
  - (b) Calculate the magnitude of the normal reaction force acting on the block. (1)
  - (c) If the acceleration of the block is  $2 \text{ m s}^{-2}$ , find the tension in the string. (3)
  - (d) If the tension in the string is 20 N, find the acceleration of the block. (2)
- (Total 7 marks)**

**Q5.**

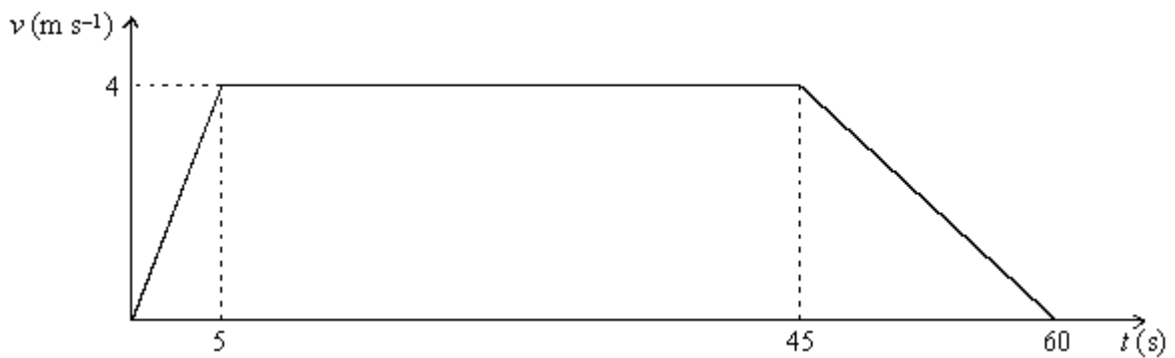
A light, inextensible string has a particle of mass 8 kg attached to one end and a particle of mass 12 kg attached to the other end. The string passes over a smooth, light pulley. The particles are released from rest with the string taut and vertical on each side of the pulley. The diagram shows the pulley and the particles.



- (a) By forming an equation of motion for each particle, show that the acceleration of the particles is  $1.96 \text{ m s}^{-2}$ . (5)
  - (b) Find the tension in the string. (2)
  - (c) Find the time that it takes for the particles to reach a speed of  $7 \text{ m s}^{-1}$ . (2)
- (Total 9 marks)**

**Q6.**

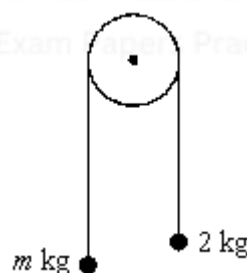
The graph shows how the velocity,  $v \text{ m s}^{-1}$ , of a cyclist varies with time,  $t$  seconds, as she moves along a straight horizontal road.



- (a) Calculate the total distance travelled by the cyclist. (3)
- (b) Find the acceleration of the cyclist during the first 5 seconds of the motion. (2)
- (c) As the cyclist moves, a force of magnitude  $P$  newtons acts on the cyclist in the direction of motion. A constant horizontal resistance force of magnitude 50 newtons also acts. Model the cyclist and her bicycle as a particle of mass 65 kg.
- (i) Draw a diagram to show **all** the forces acting on the particle. (1)
- (ii) Find the value of  $P$  during the first 5 seconds of the motion. (3)
- (Total 9 marks)**

**Q7.**

Two particles of mass 2 kg and mass  $m$  kg are connected by a light inextensible string that passes over a smooth light pulley, as shown in the diagram below.



The particles are released from rest with the string taut.

- (a) The 2 kg particle moves vertically downwards, travelling 0.8 metres in 2 seconds. Find its acceleration. (3)
- (b) Find the tension in the string. (3)

- (c) Find  $m$ , giving your answer to two significant figures.

(4)

(Total 10 marks)

**Q8.**

A block, of mass 5 kg, is held at rest on a rough plane, which is inclined at  $30^\circ$  to the horizontal. The block is released and slides down the plane. The coefficient of friction between the block and the plane is 0.2.

- (a) Draw a diagram to show the forces acting on the block as it slides. (1)
- (b) Show that the magnitude of the friction force acting on the block is approximately 8.5 N. (3)
- (c) Find the acceleration of the block. (4)
- (d) Find the speed of the block when it has travelled 1.2 metres down the slope. (3)

(Total 11 marks)

**Q9.**

A car accelerates from rest along a straight horizontal road. It experiences a constant horizontal forward force of magnitude 2000 newtons and a resistance force. The resistance force has magnitude  $40v$  newtons, when the speed of the car is  $v$  m s<sup>-1</sup>. The mass of the car is 1000 kg.

- (a) Show that

$$\frac{dv}{dt} = \frac{50 - v}{25}$$

(2)

- (b) Find the velocity of the car at time  $t$ .

(5)

(Total 7 marks)

**Q10.**

A hot air balloon is released from rest at ground level and rises vertically. It accelerates at a constant rate. It reaches a height of 120 metres after 50 seconds.

- (a) Show that the acceleration of the balloon is  $0.096 \text{ ms}^{-2}$  (3)
- (b) Find the speed of the balloon when it is at a height of 200 metres. (3)

- (c) Find the time that it takes the balloon to reach a height of 200 metres. (3)
- (d) The mass of the balloon is 250 kg. Find the magnitude of the upward force acting on the balloon. (3)
- (Total 12 marks)

**Q11.**

A van has mass 1200 kg. It travels with constant acceleration up a slope inclined at  $4^\circ$  to the horizontal. The length of the slope is 250 metres. At the bottom of the slope the van has a speed of  $20 \text{ m s}^{-1}$  and at the top its speed has dropped to  $15 \text{ m s}^{-1}$ .

- (a) Show that the acceleration of the van while it is on the slope is  $-0.35 \text{ m s}^{-2}$ . (3)
- (b) A simple model assumes that no external resistance forces act on the van. A constant force of magnitude  $P$  newtons parallel to the slope acts on the van. Find  $P$ . (4)
- (c) A more realistic model assumes that a resistance force does act on the van. This force has magnitude 300 newtons and acts parallel to the slope. Revise your answer to part (b) to take account of this extra force. (2)
- (d) In reality the resistance force will not be constant. Explain why. (2)
- (Total 11 marks)

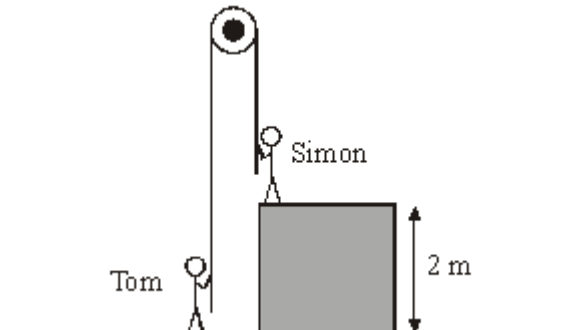
**Q12.**

A skier slides in a straight line directly down a slope inclined at  $30^\circ$  to the horizontal. The coefficient of friction between her skis and the slope is 0.3. The skier and her equipment are to be modelled as a particle of mass 80 kg. Assume that there is no air resistance present.

- (a) Draw a diagram to show the forces acting on the skier. (1)
- (b) (i) Find the magnitude of the normal reaction force acting on the skier. (2)
- (ii) Show that the magnitude of the friction force acting on the skier is 204 N to three significant figures. (2)
- (c) Find the acceleration of the skier. (4)
- (Total 9 marks)

**Q13.**

Two children are holding the ends of a light, inextensible rope, which passes over a light, smooth pulley. Initially Tom, who has a mass of 40 kg, is standing at ground level and Simon, who has a mass of 60 kg, is on the edge of a fixed platform 2 metres above ground level. Model the two boys as particles, one initially at ground level, and the other initially at a height of 2 metres. The rope is taut.



Simon steps off the platform and as he falls vertically, Tom rises vertically.

(a) Assume that the rope remains taut while the boys are moving.

(i) Show that the acceleration of each boy is  $1.96 \text{ m s}^{-2}$ .

(5)

(ii) Find the tension in the rope.

(2)

(b) Find the total distance that Tom travels upwards.

(7)

(Total 14 marks)

**Q14.**

A particle of mass  $m$  is moving along a straight horizontal line. At time  $t$  the particle has speed  $v$ . Initially the particle is at the origin and has speed  $U$ . As it moves the particle is subject to a resistance force of magnitude  $mkv^3$ .

(a) Show that  $v^2 = \frac{U^2}{2kU^2t + 1}$ .

(6)

(b) What happens to  $v$  as  $t$  increases?

(1)

(Total 7 marks)

**Q15.**

A car accelerates uniformly along a straight horizontal road for 20 seconds. In this time its speed increases from  $8 \text{ m s}^{-1}$  to  $34 \text{ m s}^{-1}$ .

(a) Calculate the acceleration of the car.

(3)

- (b) Find the distance that the car travels in this time.

(2)

- (c) The car has mass 1250 kg and a resistance force of magnitude 400 N acts on the car. Find the magnitude of the forward driving force that acts on the car.

(3)

(Total 8 marks)

### Q16.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

- (a) Find the resultant force on the particle.

(2)

- (b) The mass of the particle is 8 kg. Find the acceleration of the particle.

(1)

- (c) At time  $t = 0$ , the particle is at the origin and has velocity  $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$ .

- (i) Find the velocity of the particle when  $t = 40$  seconds.

(2)

- (ii) Find the distance of the particle from the origin when  $t = 40$  seconds.

(5)

(Total 10 marks)

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### Q17.

A rope is used to pull a box, of mass 20 kg, up a rough slope. Assume that the box moves with a constant acceleration.

- (a) As the box moves 4 metres up the slope its speed increases from  $1 \text{ m s}^{-1}$  to  $5 \text{ m s}^{-1}$ . Calculate the acceleration of the box.

(3)

- (b) The slope is at an angle of  $30^\circ$  to the horizontal. Assume that the rope is parallel to the slope. The coefficient of friction between the box and the slope is 0.4.

- (i) Draw a diagram to show the forces on the box.

(1)

- (ii) Show that the magnitude of the friction force acting on the box is approximately 68 N.

(3)



- (iii) Find the tension in the rope.

(4)

(Total 11 marks)

**Q18.**

A car accelerates uniformly along a straight horizontal road. As it travels 200 metres, its speed increases from  $12 \text{ m s}^{-1}$  to  $20 \text{ m s}^{-1}$ .

- (a) Show that the acceleration of the car is  $0.64 \text{ m s}^{-2}$ .

(3)

- (b) Find the time that it takes the car to travel this distance.

(3)

- (c) The mass of the car is 1200 kg. Using the answer from part (a), find the magnitude of the forward force that acts on the car during the above motion if:

- (i) there is not resistance to its motion;

(1)

- (ii) a constant resistive force of magnitude 450 newtons acts on the car.

(2)

(Total 9 marks)

**Q19.**

A lift of mass 600 kg is suspended from a single vertical cable. The lift is initially at rest and then starts to accelerate upwards. The tension in the cable remains constant at 6060 newtons while the lift is accelerating. Assume that there is no resistance to the motion of the lift.

- (a) Show that the acceleration of the lift is  $0.3 \text{ m s}^{-2}$ .

(2)

- (b) Find the time that it would take the lift to travel 8 metres.

(3)

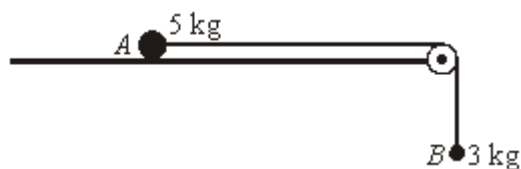
- (c) Find the speed of the lift when it has travelled 8 metres.

(2)

(Total 7 marks)

**Q20.**

The diagram shows two particles that are connected by a light inextensible string. The 5 kg particle, *A*, is on a rough horizontal surface and the 3 kg particle, *B*, hangs on the other end of the string as shown in the diagram. The string passes over a smooth light pulley.



The coefficient of friction between  $A$  and the horizontal surface is 0.8. The particles are set in motion with the string taut and  $A$  moving towards the pulley. The initial speed of both particles is  $2 \text{ m s}^{-1}$ . The particles slow down and stop before  $A$  reaches the pulley or  $B$  hits the floor.

- (a) Show that, while both particles are moving, the magnitude of their acceleration is  $1.225 \text{ m s}^{-2}$ .

(6)

- (b) Find the tension in the string while the particles are moving.

(2)

- (c) Find the distance that the particles move before they come to rest.

(3)

(Total 11 marks)



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